

To study the variation of cell potential in $\text{Zn} \mid \text{Zn}^{2+} \parallel \text{Cu}^{2+} \mid \text{Cu}$ with change in concentration of electrolytes (CuSO_4 or ZnSO_4) at room temperature

Video Reference :- <https://www.youtube.com/watch?v=9wgSwoJZA0w>

Materials Required :-

- Zinc Strip
- Copper Strip
- 1 Molar Zinc Sulphate solution
- Voltmeter
- 1 Molar Copper Sulphate solution
- Salt Bridge
- Connecting Wires
- Sand Paper

Procedure :-

- Take the copper strip & clean it with the help of sandpaper.
- Dip the copper strip into the beaker containing 1 Molar copper sulphate solution.
- Take the zinc strip & clean it with sandpaper.
- Dip it into another beaker containing 1 Molar Zinc Sulphate solution.
- Take the salt bridge & connect the two solutions through the salt bridge.
- Connect the positive terminal of the voltmeter with the copper strip using connecting wire.
- Connect the negative terminal of the voltmeter with the zinc strip using another connecting wire.
- Note the position of the pointer in the voltmeter & record the reading.
- Repeat the experiment by taking the different concentrations of zinc sulphate & copper sulphate solutions.

Concentration of ZnSO_4 Solution	Concentration of CuSO_4 Solution	E.M.F of the cell
1 M	1 M	1.1 V
1 M	0.5 M	1.091 V
1 M	0.025 M	1.052 V
1 M	0.0125 M	1.043 V
0.5 M	1 M	1.108 V
0.025 M	1 M	1.147 V
0.0125 M	1 M	1.156V